

If $x_1 = [a_1 \ b_1 \ c_1]; x_2 = [a_2 \ b_2 \ c_2]; x_3 = [a_3 \ b_3 \ c_3]$, then the vectors x_1, x_2, x_3 are said to be linearly dependent if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

- (2) The combination $k_1x_1 + k_2x_2 + \dots + k_nx_n$ is said to be linearly independent if $k_1x_1 + k_2x_2 + \dots + k_nx_n = 0$ when $k_1 = k_2 = k_3 = \dots = k_n = 0$

2.14.1 Eigen Values and Eigen Vectors

For any square Matrix $A_{n \times n}$, the equation $|A - \lambda I| = 0$ where ' λ ' is a scalar is called the characteristic equation.

The roots of the characteristic equation of a Matrix are called Eigen Values.

2.14.2 Properties of Eigen Values

- (i) If $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ are 'n' Eigen Values of $A_{n \times n}$, then
 - (a) Sum of Eigen Values of 'A' = $\lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_n = \sum_{i=1}^n \lambda_i = \text{trace}(A)$ = Sum of Principal diagonal elements
 - (b) Product of all the Eigen Values of 'A' = $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 \cdot \dots \cdot \lambda_n = \prod_{i=1}^n \lambda_i = |A|$
 - (c) Eigen Values of A^m are $\lambda_1^m, \lambda_2^m, \lambda_3^m, \dots, \lambda_n^m$
 - (d) Eigen Values of $\text{adj}(A)$ are $\frac{|A|}{\lambda_1}, \frac{|A|}{\lambda_2}, \frac{|A|}{\lambda_3}, \dots, \frac{|A|}{\lambda_n}$
 - (e) Eigen Values of A & A^T are the same.
 - (f) Eigen Values of $k_1A + k_2I$ (Where k_1 and k_2 are scalar) are $k_1\lambda_1 + k_2, k_1\lambda_2 + k_2, k_1\lambda_3 + k_2, k_1\lambda_4 + k_2, \dots, k_1\lambda_n + k_2$
- (ii) '0' is always an Eigen Value of an odd-order Skew-Symmetric Matrix.
- (iii) Eigen Values of a Real Symmetric Matrix are always real.
- (iv) Eigen Values of the Skew-Symmetric Matrix are either zero (or) purely Imaginary.
- (v) The Eigen values of an Orthogonal Matrix are of unit modulus.
- (vi) If the sum of all the elements in a row (or Column) is constant (= k) for all the rows (or columns) in the matrix respectively, then 'k' is an Eigen Value of the Matrix.

Example: If $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ and if $a_1 + b_1 + c_1 = a_2 + b_2 + c_2 = a_3 + b_3 + c_3 = k$,

then 'k' is an Eigen Value of 'A'.

- (vii) The Eigen Values of an upper triangular Matrix, a lower triangular Matrix, a diagonal Matrix are the Principal diagonal elements of the Matrix.

2.15 Eigen Vector

A non-zero column vector $X_{n \times 1}$ is said to be an Eigen Vector of $A_{n \times n}$ corresponding to the Eigen Value ' λ ', if $AX = \lambda X (X \neq 0)$.

2.15.1 Properties of Eigen Vectors

- (i) Eigen Vectors of A & A^T are not the same.
- (ii) Eigen Vectors of A & A^M are same.
- (iii) The Eigen Vectors of a Real Symmetric Matrix are always orthogonal.
- (iv) The number of linearly independent Eigen Vectors of ' $A_{n \times n}$ ' is equal to the number of distinct Eigen Values of ' $A_{n \times n}$ '.

2.15.2 Cayley Hamilton Theorem

Every Matrix satisfies its characteristic equation.

This means that, if $c_0\lambda^n + c_1\lambda^{n-1} + \dots + c_{n-1}\lambda + c_n = 0$ is the characteristic equation of a square matrix A of order n , then

$$c_0A^n + c_1A^{n-1} + \dots + c_{n-1}A + c_nI = 0 \quad \dots(i)$$

Note: When λ is replaced by A in the characteristic equation, the constant term c_n should be replaced by c_nI to get the result of the Cayley-Hamilton theorem, where I is the unit matrix of order n .

Also, 0 in the R.H.S. of (i) is a null matrix of order n .

2.16. Subspace (Basis of Dimensions)

2.16.1 Vector

An ordered n -tuple of numbers is called an n -vector.

2.16.2 Linearly Independent and Dependent Vector

Let X_1 and X_2 be the non-zero vectors:

- $\{x_1, x_2, \dots, x_k\}$ are linearly independent if $r_1x_1 + r_2x_2 + \dots + r_kx_k = 0$ only for $r_1 = r_2 = \dots = r_k = 0$.
- The vectors $x_1, r_2, \dots, x_k$ are linearly dependent if they are not linearly independent; that is, if there exist scalars $r_1, r_2, \dots, r_k$ which are not all zero such that

$$r_1x_1 + r_2x_2 + \dots + r_kx_k = 0$$

Note: Let $X_1, X_2, \dots, X_n$ be ' n ' vector of matrix A .

- If $\text{rank}(A) = \text{number of vectors}$ then vector $X_1, X_2, \dots, X_n$ are linearly independent.
- If $\text{rank}(A) \neq \text{number of vectors}$ then vector $X_1, X_2, \dots, X_n$ are linearly dependent.

2.16.3 Vector Space $\mathbb{R}^n$

If n positive integer, then an ordered n -tuple is a sequence of n real numbers $(\alpha_1, \alpha_2, \dots, \alpha_n)$. the set of all ordered n -tuples is called n -space and is denoted by $\mathbb{R}^n$.

2.16.4 Subspaces of an N -vector space V_n

A non-empty set S , of vectors of $V_n(F)$, is called a subspace of $V_n(F)$, if

- ξ_1, ξ_2 are any two members of S , then $\xi_1 + \xi_2$ is also a member of S ; and
- ξ is a member of S , and k is a scalar then $k\xi$ is also a member of S .

Briefly, we may say that a set S of vectors $V_n(F)$ is a subspace of $V_n(F)$ if closed w.r.t. the compositions of “addition” and “multiplication with scalars”.

Every subspace of V_n contains the zero vector; being the product of any vector with the scalar zero.

2.16.5 Construction of Subspaces

- **A subspace Spanned by a Set of Vectors:** A subspace that arises as a set of all linear combinations of any given set of vectors is said to be spanned by the given set of vectors.
- **Basis of a subspace:** A set of vectors is said to be a basis of a subspace, if
 - The subspace is spanned by the set, and
 - The set is linearly independent.

Note: If we have N vectors and they are independent then they span N -dimension space. But if they are dependent then they span only a subspace of N -dimension space.

2.16.6 Orthogonality of Vectors

- Two vectors are orthogonal if each is non-zero and $X_1^T X_2 = 0$
- If n vectors $X_1, X_2, \dots, X_n$ each of n dimensions is orthogonal then they are surely linearly independent and form the basis for n -dimension space.
- The set of the vector is orthonormal if they are orthogonal and have unit magnitude.

2.17 Similar Matrices

- Two matrix A and B are similar if there exist a non singular matrix P such that $B = P^{-1}AP$
- Similar matrix has same eigen values
- If A is similar to B then B is also similar to A
- If A is similar to B and B is similar to C then A is similar to C .

2.18 Diagonalization of a matrix

Finding a matrix D which is a diagonal matrix and which is similar to A is called diagonalization i.e., we wish to find a non-singular matrix M such that $A = M^{-1}DM$ where D is a diagonal matrix.

2.18.1 Condition for a Matrix to be Diagonalizable

1. A necessary and sufficient condition for a matrix $A_{n \times n}$ to be diagonalizable is that the matrix must have n linearly independent eigen vectors.
2. A sufficient (but not necessary) condition for a matrix $A_{n \times n}$ to be diagonalizable is that the matrix must have n linearly independent eigen values.

This is because if a matrix has n linearly independent eigen values then it surely has n linearly independent eigen vectors (although the converse of this is not true).

